

Matthew N. Mastroeni

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Employment

Assistant Professor , SUNY Polytechnic Institute	August 2024 – Present
Postdoc Research Associate , Iowa State University	August 2021 – May 2024
Postdoctoral Fellow , Oklahoma State University	August 2018 – May 2021

Education

Ph.D. in Mathematics , University of Illinois at Urbana-Champaign Thesis: <i>Betti Numbers of Koszul Algebras and Codimension Two Matrix Factorizations</i> Advisor: Hal Schenck	2018
M.S. in Mathematics , Syracuse University	2012
B.A. in Mathematics , Ithaca College	2009

Research

Interests: Computational, combinatorial, and homological aspects of commutative algebra, especially the structure of free resolutions, Koszul algebras, linkage, and Rees algebras.

Publications:

* = undergraduate coauthor

- (1) A. LaClair, M. Mastroeni, J. McCullough, and I. Peeva. *Koszul binomial edge ideals*. Submitted.
- (2) A. LaClair, M. Mastroeni, J. McCullough, and I. Peeva. *Koszul graded Möbius algebras and strongly chordal graphs*. Sel. Math. New Ser. 31, 29 (2025).
- (3) M. Mastroeni, J. McCullough, A. Osborne*, J. Rice, and C. Willis*. *Depth and singular varieties of exterior edge ideals*. To appear in Rocky Mountain J. Math. [arXiv:2208.03366](https://arxiv.org/abs/2208.03366)
- (4) M. Mastroeni and J. McCullough. *Chow rings of matroids are Koszul*. Math. Ann. 387 (2023), no.3-4, 1819–1851.

- (5) C. Francisco, M. Mastroeni, J. Mermin, and J. Schweig. *Computing generalized Frobenius powers of monomial ideals*. Submitted. [arXiv:2005.14643](#)
- (6) L. Ferraro, F. Galetto, F. Gandini, H. Huang, M. Mastroeni, and X. Ni. *The InvariantRing package for Macaulay2*. J. Softw. Algebra Geom. 14 (2024), 5–11.
- (7) P. Mantero and M. Mastroeni. *The structure of Koszul algebras defined by four quadrics*. J. Algebra 601 (2022), 280–311.
- (8) M. Mastroeni, H. Schenck, and M. Stillman. *Quadratic Gorenstein rings and the Koszul property II*. Int. Math. Res. Not. IMRN 2023, no. 2, 1461–1482.
- (9) P. Mantero and M. Mastroeni. *Betti numbers of Koszul algebras defined by four quadrics*. J. Pure Appl. Algebra 225 (2021), no. 2, Paper No. 106504.
- (10) M. Mastroeni, H. Schenck, and M. Stillman. *Quadratic Gorenstein rings and the Koszul property I*. Trans. Amer. Math. Soc. 374 (2021), no. 2, 1077–1093.
- (11) M. Mastroeni. *Koszul almost complete intersections*. J. Algebra 501 (2018), 285–302.
- (12) M. Mastroeni. *Matrix factorizations and singularity categories in codimension two*. Proc. Amer. Math. Soc. 136 (2018), no. 11, 4605–4617.

Honors and Awards

Postdoctoral Scholar Excellence Award for Teaching/Mentoring

2022

Iowa State University

One of only two university-wide awards based on a teaching statement, CV, student evaluations, and classroom observation.

AMS-Simons Travel Grant

2021

Departmental TA Instructional Award

2017

University of Illinois Math Department

Annual departmental award based on a teaching portfolio, student evaluations, an interview, and classroom observation.

REGS Day Award

2013

University of Illinois Math Department

Award for the best summer research project on higher codimension matrix factorizations.

Donald E. Kibbey Prize

2010

Syracuse University Math Department

University Fellowship

2009

Syracuse University

Teaching Experience

Instructor (SUNY Polytechnic Institute)

MAT 151 Calculus 1	Spring 2025, Fall 2024
MAT 152 Calculus 2	Fall 2025
MAT 280 Linear Algebra	Spring 2025, Fall 2024
MAT 381 Modern Algebra	Fall 2025

Large Lecture Instructor (Iowa State University)

Math 151 Calculus for Business and Social Sciences	Fall 2021
Math 165 Calculus 1	Spring 2023, Spring 2022
Math 267 Elementary Differential Equations and Laplace Transforms	Spring 2024

Instructor (Oklahoma State University)

Math 2144 Calculus 1	Spring 2021, Fall 2020, Fall 2018
Math 2153 Calculus 2	Fall 2019
Math 3013 Linear Algebra	Spring 2020
Math 3613 Intro to Abstract Algebra	Spring 2020, Fall 2019, Spring 2019

Instructor (University of Illinois)

Math 124 Finite Mathematics	Spring 2018
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Instructor (Syracuse University)

MAT 221 Elementary Probability and Statistics 1	Fall 2011, Spring 2011
MAT 286 Calculus for the Life Sciences	Spring 2012

Recitation Instructor (Iowa State University)

Math 166 Calculus 2	Fall 2023, Fall 2022
Math 267 Elementary Differential Equations and Laplace Transforms	Fall 2021

Teaching Assistant (University of Illinois)

Math 221 Calculus 1	Fall 2016, Fall 2015
Math 231 Calculus 2	Fall 2017, Spring 2016, Spring 2015, Fall 2014, Spring 2014, Spring 2013, Fall 2012
Math 241 Calculus 3	Fall 2013

Teaching Assistant (Syracuse University)

MAT 121 Probability and Statistics for the Liberal Arts 1

Fall 2010

Grader (University of Illinois)

Math 416 Abstract Linear Algebra

Spring 2018

Mentoring

Undergraduate Research Advising:

* = produced undergraduate coauthored publication

► Connor Donovan

Spring 2025

Combinatorics of licci Artinian monomial ideals
SUNY Polytechnic Institute

► Matthew Dorang and Kelvin Ma

Spring 2024

Homologically extreme quadratic monomial ideals coming from Coxeter groups
Iowa State University

► Andrew Osborne and Cole Willis*

Spring 2022

Depth and singular varieties of exterior edge ideals
Iowa State University

► Paul Genelly, Advith Govindarajan, Lang Yin, and Yuan Zhang

Summer 2017

A game with Noetherian rings
University of Illinois

Invited Talks

AMS Special Session on Commutative Algebra

October 2025

St. Louis University

Route 81 Conference on Commutative Algebra and Algebraic Geometry
invited plenary talk

November 2024

Cornell University

AMS Special Session on Commutative Algebra and Its Connections to Combinatorics

September 2024

San Antonio, TX

AMS Special Session on Homological Methods in Commutative Algebra and Algebraic Geometry San Francisco, CA	May 2024
AMS Special Session on Connections between Commutative Algebra and Algebraic Combinatorics Milwaukee, WI	April 2024
AMS Special Session on Group Actions in Commutative Algebra San Francisco, CA	January 2024
Algebra Seminar New Mexico State University	November 2023
AMS Special Session on Commutative Algebra Omaha, NE	October 2023
Commutative Algebra and Algebraic Geometry Seminar University of Minnesota	April 2023
Fellowship of the Ring National Seminar virtual	January 2023
AMS Special Session on Topological and Combinatorial Methods in Commutative Algebra Boston, MA	January 2023
AMS Special Session on Interactions between Combinatorics and Commutative Algebra El Paso, TX	September 2022
AMS Special Session on Commutative Algebra virtual	May 2022
Algebra and Geometry Seminar Queen's University	April 2022
Algebra Seminar University of Arkansas	March 2022
AMS Special Session on Combinatorial Methods in Commutative Algebra virtual	March 2022
CA+ Conference invited plenary talk Minneapolis, MN	October 2021
AMS Special Session on Commutative Algebra virtual	October 2021
AMS Special Session on Commutative Algebra virtual	April 2021

AMS Special Session on Commutative Algebra and its Interaction with Algebraic Geometry and Combinatorics virtual	March 2021
Algebra and Geometry Seminar Iowa State University	October 2020
AMS Special Session on Commutative Algebra and Connections to Algebraic Geometry and Combinatorics virtual	October 2020
AMS Special Session on Combinatorial Techniques in Commutative Algebra West Lafayette, IN	April 2020 (canceled)
Algebra Seminar University of Arkansas	April 2019
AMS Special Session on Interactions Between Combinatorics and Commutative Algebra Fayetteville, AR	November 2018
AMS Special Session on Commutative Algebra and Complexity Ann Arbor, MI	October 2018
Math Department Colloquium Oklahoma State University	October 2018
Algebra Seminar University of Arkansas	April 2018
Combinatorial and Commutative Algebra Seminar Oklahoma State University	April 2018
Commutative Algebra Seminar University of Nebraska–Lincoln	April 2018
AMS Special Session on Commutative and Combinatorial Algebra Columbus, OH	March 2018
AMS Special Session on Commutative Algebra in All Characteristics San Diego, CA	January 2018
Structures on Free Resolutions Conference Lubbock, TX	October 2017

Institutional Service and Outreach

Putnam Exam Advisor SUNY Poly Math Department	Fall 2025
Open House Math Department Recruitment SUNY Polytechnic Institute	Fall 2024
Graduate Affairs Committee Member University of Illinois Math Department	Fall 2017 – Spring 2018
TA Teaching Awards Committee Member University of Illinois Math Department	Fall 2017
Commutative Ring Theory Seminar Organizer University of Illinois	Fall 2014 – Spring 2018

Professional Service and Outreach

Arizona-New Mexico Symposium on Commutative Algebra and its Interactions: Geometric Combinatorics <i>Invited research project co-leader.</i>	December 2025
University of Arkansas Spring Lecture Series Panel <i>Invited organizer and moderator of a career advice panel for grad students.</i>	May 2025
Macaulay2 Workshop Project Co-Leader Tulane University	April 2025
AWM Graduate Student Poster Session Judge Joint Math Meetings	January 2023
Stillwater High School Math Seminar Stillwater, OK	April 2020 (canceled), November 2019
NSF Grant Reviewer	1 time
Referee	

For journals/conference proceedings including:

- Algebra and Number Theory
- Transactions of the AMS
- International Journal of Algebra and Computation
- Journal of Commutative Algebra
- Journal of Software for Algebra and Geometry
- Springer AWM series

Software Creation

LatticeChowRings package for Macaulay2

2022

<https://github.com/mnmastro/LatticeChowRings>

A package for working with the Feichtner-Yuzvinsky Chow rings of atomic lattices with respect to a given building set, including the augmented Chow rings and graded Möbius algebras of matroids.

InvariantRing package for Macaulay2

2020

with L. Ferraro, F. Galetto, F. Gandini, T. Hawes, H. Huang, and X. Ni

<https://github.com/galettof/InvariantRing>

A package for computing invariants of group actions on polynomial rings. Includes a variety of methods for computing invariants of finite groups, diagonal actions of tori and finite abelian groups, and actions of linearly reductive groups.

TestIdeals package for Macaulay2

2019

with E. Bela, A. Boix, J. Bruce, D. Ellingson, D. Hernández, Z. Kadyrsizova, M. Katzman, S. Malec, M. Mostafazadehfard, M. Robinson, K. Schwede, D. Smolkin, P. Teixeira, and E. Witt

A package for working with singularities in positive characteristic via computations of test ideals and related objects.

Professional Development

Algebra Days Workshop

November 2023

Arizona State University

Macaulay2 Workshop

May 2020

Cleveland State University (online workshop)

CIME Recent Developments in Commutative Algebra Workshop

July 2019

CBMS Conference on Applications of Polynomial Systems Texas Christian University	June 2018
Macaulay2 Workshop University of Wisconsin Madison	April 2018
MSRI Homological Conjectures Workshop	March 2018
Stillman's Conjecture and other Progress on Free Resolutions UC Berkeley	July 2017
RTG Homological Conjectures in Commutative Algebra Workshop University of Illinois Chicago	November 2016
Macaulay2 Workshop University of Utah	May 2016
RTG Local Cohomology Workshop University of Illinois Chicago	February 2015